

A MAJOR PROJECT PHASE II REPORT

On

"Design and Implementation of Deterministic AXI Based Direct Memory Access Controller with Bandwidth Reservation and Worst Case Execution Time."

Submitted in partial fulfillment of the requirement for the award of the degree Bachelor of Engineering

In

DEPARTMENT OF ELECTRONICS ENGINEERING

(VLSI DESIGN AND TECHNOLOGY)

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CERTIFICATE

This is to Certified that the Major Project Phase II work titled "**Design and Implementation of Deterministic AXI Based Direct Memory Access Controller with Bandwidth Reservation and Worst Case Execution Time"** is carried out by **CHANNAYYA METI (1MJ23VL004), HARISH RAGHUNATHAN (1MJ23VL007),PRASHANTH REDDY (1MJ23VL014),ANKUSH S (1MJ24VL401)** who are bonafide students of MVJ College of Engineering, Bengaluru, in partial fulfillment for the award of degree of **Bachelor of Engineering in Electronics Engineering(VLSI DESIGN AND TECHNOLOGY)**of the Visvesvaraya Technological University, Belagavi during the year 2026-2027. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project phase II report deposited in the departmental library. The major project report II has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said Degree.

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DECLARATION

We, CHANNAYYA METI (1MJ23VL004), HARISH RAGHUNATHAN (1MJ23VL007),PRASHANTH REDDY (1MJ23VL014),ANKUSH S (1MJ24VL401) students of Seventh semester B.E., **Department of Electronics Engineering(VLSI DESIGN AND TECHNOLOGY)**, MVJ College of Engineering, Bengaluru, hereby declare that the major project phase II titled **"Design and Implementation of Deterministic AXI Based Direct Memory Access Controller with Bandwidth Reservation and Worst Case Execution Time"**has been carried out by us and submitted in partial fulfilment for the award of Degree of **Bachelor of Engineering** in **Electronics Engineering(VLSI DESIGN AND TECHNOLOGY)**during the year 2026-2025.Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and we will be one of the authors of the same.

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Thank You

ABSTRACT

The Advanced eXtensible Interface (AXI) protocol is widely used in FPGA-based systems for high-speed data communication between processing units, memory, and peripheral devices. However, conventional AXI-based interconnects may experience unpredictable transfer delays, bus contention, and variations in data-transfer performance when multiple masters compete for shared resources. These challenges highlight the need for deterministic data transfer, efficient bandwidth utilization, and improved timing predictability in real-time applications.

This project presents the design and implementation of a deterministic AXI-based Direct Memory Access (DMA) controller with bandwidth reservation and Worst-Case Execution Time (WCET) monitoring. The implemented system to improve data-transfer predictability by incorporating a Token Bucket mechanism for bandwidth management and Time Division Multiple Access (TDMA) scheduling for controlled access to shared resources. A WCET monitoring mechanism is included to observe transfer execution time and identify timing-limit violations. The design is developed using hardware description language concepts and evaluated through RTL simulation and FPGA design tools.

The project aims to provide more predictable bus access, fair resource sharing, and improved suitability for real-time FPGA-based applications. The design demonstrates an approach to integrating bandwidth management and WCET monitoring into an AXI-based DMA controller.

Keywords: AXI Protocol, Direct Memory Access (DMA), Bandwidth Reservation, Token Bucket, TDMA Scheduling, Worst-Case Execution Time (WCET), FPGA, RTL Simulation.

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CHAPTER 1

INTRODUCTION

INTRODUCTION

The rapid development of digital systems has increased the need for reliable and efficient communication between processors, memory units, and peripheral devices. In modern embedded and FPGA-based systems, large amounts of data must be transferred while making effective use of available hardware resources. The communication interface therefore plays an important role in determining overall system performance, transfer latency, and the ability of a design to meet timing requirements.

The Advanced eXtensible Interface (AXI) is part of the Advanced Microcontroller Bus Architecture (AMBA) family and is widely used to connect functional blocks in System-on-Chip (SoC) and FPGA designs. AXI supports structured communication through independent channels and a VALID–READY handshaking mechanism. These features allow connected components to coordinate data transfers. However, the performance of a complete system also depends on the interconnect structure, arbitration policy, traffic pattern, and availability of shared resources.

Direct Memory Access (DMA) is a technique that allows data to be moved between memory and peripherals without requiring the processor to manage every transfer operation. A DMA controller can reduce processor involvement and make data movement more efficient. In systems with more than one requesting master, access to shared resources must be managed carefully. If requests arrive at the same time, arbitration and scheduling determine which master is served and how long other masters must wait.

This project focuses on the design and implementation of a deterministic AXI-based Direct Memory Access controller with bandwidth reservation and Worst-Case Execution Time (WCET) monitoring. The proposed design considers Token Bucket-based bandwidth management and Time Division Multiple Access (TDMA) scheduling to organize access to shared resources. A WCET monitoring mechanism is included to observe transfer execution time against a defined timing limit. The design is examined using RTL simulation and FPGA design tools, with attention to master selection, transfer activity, timing reports, and reported power information.

The purpose of the project is to investigate how bandwidth management and execution-time monitoring can be combined in an AXI-based DMA design. Rather than treating throughput as the only objective, the project also considers predictability and fairness when multiple masters compete for a shared interface. The following sections explain the background, motivation, challenges, objectives, and scope of the proposed work.

1.1 Background of AXI-Based DMA Systems:

An AXI-based system contains components that initiate transactions and components that respond to them. A master initiates a request, while a slave or target responds to the transaction. In a design containing multiple masters, an interconnect or arbitration mechanism is required to coordinate access to common targets. AXI uses handshaking signals to indicate when information is valid and when the receiving component is ready to accept it. A transfer takes place when the corresponding VALID and READY signals are asserted together.

The handshaking mechanism is useful because connected components may operate at different rates. A sender can indicate that information is available, and the receiver can indicate when it is able to accept that information. Nevertheless, handshaking by itself does not ensure that every master receives a fixed share of the available bandwidth or that every transaction completes within a known time bound. These properties depend on scheduling logic, buffering, target response, and assumptions about the surrounding system.

A DMA controller coordinates data movement while reducing the need for repeated processor intervention. In an AXI-based design, the controller must manage requests, data availability, transfer acceptance, and completion conditions. When several masters are present, the design must also decide how competing requests are handled. An arbitration policy that responds only to the current request state may produce different waiting times under different traffic patterns.

Deterministic behaviour is important when a system must plan its activity around timing requirements. In this context, determinism means that transfer behaviour is governed by defined rules and assumptions, rather than being left entirely to unpredictable competition for resources. A deterministic schedule can make expected access opportunities clearer. A strict execution-time bound, however, also depends on

implementation details, clocking, response delays, and the assumptions used in timing analysis.

The project studies a controller that combines access scheduling, bandwidth management, and execution-time observation. The signals identified in the project presentation include clock and reset inputs, request and ready signals for two masters, memory-side transfer signals, a current-master indication, and a WCET counter. These signals provide a basis for observing which master is selected, whether a transfer is accepted, and how the execution-time counter changes during simulation.

1.2 Need for Deterministic Data Transfer

Shared communication resources can become a performance limitation when multiple masters request service concurrently. If the arbitration policy does not provide an explicit reservation or bounded scheduling rule, the time spent waiting for access may change with the amount and ordering of traffic. This variation can complicate system-level planning, especially where data movement must occur within a specified time interval.

Bandwidth reservation provides a way to manage how communication capacity is allocated. Instead of allowing all requests to compete without an explicit allocation policy, a controller can use rules that limit or organize access for each requester. The project considers a Token Bucket mechanism as a way of representing or controlling permitted traffic. In a complete implementation, the bucket capacity, token-generation rate, and conditions for consuming tokens must be defined so that the mechanism has a clear and testable meaning.

TDMA scheduling divides access opportunities into time slots. A schedule can assign different slots to different masters, giving each master a planned opportunity to use a shared resource. The project presentation describes fixed TDMA slots. Slot duration, master assignment, idle-slot behaviour, and handling of a master that has no pending request are important design choices and should be confirmed in RTL and simulation.

WCET monitoring addresses a related but distinct requirement. Bandwidth reservation and scheduling regulate access, whereas WCET monitoring observes elapsed execution cycles or time and checks that activity against a configured limit. Monitoring alone does not prove that a system always meets a worst-case deadline.

Combining these mechanisms can provide a structured approach to managing competing requests. The schedule establishes access opportunities, bandwidth management constrains or accounts for traffic, and the counter observes transfer duration. The effectiveness of this combination must be assessed from the implemented logic and test results rather than assumed in advance

1.3 Existing Challenges in AXI Interconnects and DMA

Existing AXI interconnect designs use a variety of approaches to arbitrate requests and improve communication efficiency. The literature listed in the project presentation covers hybrid interconnect structures, AXI4-Stream interconnects, VALID–READY handshake mechanisms, counter-based approaches, FPGA bus interfaces, high-speed memory interfaces, lightweight DMA designs, and low-power AXI implementations. These works show that interconnect architecture, arbitration, throughput, resource utilization, and power are recurring concerns in AXI-based systems.

One challenge is variable waiting time. When multiple masters request access, the arbitration scheme and the duration of previous transactions can influence when a waiting master is served. A scheme designed to improve average throughput does not automatically provide a bound on maximum waiting time. The scheduling policy must therefore be selected with the intended timing requirements in mind.

A second challenge is contention for shared resources. If several requesters compete for the same memory or interface, the design must coordinate access while preserving correct handshaking. Incorrect handling of VALID and READY signals can lead to stalled or incorrectly accepted transfers. Consequently, the controller's state transitions and handshake conditions must be verified in simulation.

A further challenge is the absence of an explicit bandwidth allocation policy. A conventional arbitration mechanism may select a master based on priority or request order without enforcing a specified bandwidth share. If the application requires predictable allocation, the design needs a reservation or scheduling rule and a way to verify that the rule is followed.It is important to distinguish performance measurements from guaranteed properties. A simulation waveform demonstrates behaviour for the inputs tested, while timing analysis reports whether the implemented logic meets the timing constraints provided to the tool. Neither alone proves that every possible

transaction will meet a worst-case execution deadline. Claims about WCET, starvation prevention, or percentage improvements should be supported by a defined methodology, assumptions, and comparison baseline.

Finally, FPGA designs must balance functionality with hardware resources and power. Additional counters, scheduling logic, and traffic-control mechanisms may improve observability or predictability but can also add logic and switching activity. The implementation should therefore be evaluated using functional evidence and the available timing, resource, and power reports.

1.4 Bandwidth Reservation and TDMA Scheduling

Bandwidth reservation is intended to make resource allocation more structured by defining how communication capacity is allocated. In the proposed project, Token Bucket-based management is identified as one component. A Token Bucket model generally uses a token capacity and a token-generation rule to represent an allowed rate of traffic. A request or transfer may be permitted when sufficient tokens are available, depending on the implemented policy. The exact role of the bucket in this project should match the RTL and be documented with its parameters.

TDMA is a scheduling method in which access to a shared resource is divided into time slots. A fixed schedule can associate particular slots with particular masters, so the opportunity to access the resource is determined by slot position. This can make service opportunities easier to anticipate than a policy that selects a master only through instantaneous priority competition. Actual timing properties still depend on slot length, transfer protocol, target response, and behaviour when a selected master is not ready.

The project presentation describes fixed eight-cycle TDMA slots. If this is the configuration used in the final RTL, the report should explain how the eight-cycle interval is counted, how master identities are assigned to slots, and what happens when a request is absent or a transfer cannot complete. Simulation should confirm that the observed master-selection sequence agrees with the intended schedule. Slot length should not be treated as a guaranteed transaction completion time unless all transfer and response conditions are bounded.

Token Bucket and TDMA mechanisms address related but different aspects of traffic control. Token Bucket logic can represent a traffic allowance or rate policy, while

TDMA determines when a master receives an access opportunity. Using both may provide complementary control, but their interaction must be explicitly defined. For example, the design must specify whether a scheduled master without sufficient tokens can use its slot, whether unused slots remain idle or are reassigned, and how counters are updated during stalls.

Test cases should include simultaneous requests, a single active master, alternating requests, idle intervals, reset behaviour, and transfers delayed by the receiving side. The resulting waveforms can then be used to evaluate whether the scheduling and bandwidth-control logic follows the documented rules.

1.5 Worst-Case Execution Time Monitoring

Worst-Case Execution Time refers to an upper bound on the time required for an operation to complete under a defined set of operating conditions and assumptions. In a hardware controller, execution time may be measured in clock cycles or converted into time using the clock period. Monitoring this quantity can help identify transfers that take longer than an expected or configured limit. It is relevant when a design is intended to support applications with timing constraints.

In the proposed AXI-based DMA controller, a WCET counter is included among the signals identified in the project presentation. Such a counter can track elapsed cycles during a defined transfer or monitoring interval. To interpret the counter correctly, the implementation must specify when counting begins, when it stops, whether it resets between transfers, how stalls are handled, and what action occurs if the limit is reached.

A stronger claim—that every transfer completes within a worst-case bound—requires additional evidence. The bound must account for arbitration, TDMA slot position, handshake stalls, memory or slave response time, and any other delay included in the system model. If these factors are not bounded, the counter can report observed duration but it cannot by itself establish a universal execution-time guarantee.

FPGA timing reports provide a separate view of implementation timing. Positive worst negative slack indicates that the reported setup paths meet the specified timing requirement under the analysed constraints. Hold slack and failing-endpoint information provide additional checks.

1.6 Objectives

- The primary objective of this project is to develop a deterministic AXI-based Direct Memory Access controller that guarantees predictable data transfers.
- To implement on Bandwidth reservation using Token Bucket and Time Division Multiple Access scheduling methods to control bus access.
- And testing it with Worst Case Execution Time analysis is carried out to calculate maximum possible delays. Finally, the architecture is implemented and tested on an FPGA platform to validate performance and timing behavior.

CHAPTER 2

LITERATURE SURVEY

LITERATURE SURVEY

2.1 Hybrid AXI Bus Interconnect Structure Design for FPGA-Based Image Processing Systems

Authors: Youyao Liu, Wanyu Bai, Zhihong Huang, Gang Cai
Publication: AIPR Conference, 2024

The authors proposed a hybrid AXI bus interconnect for FPGA-based image processing systems. The study focuses on improving communication between processing modules and memory units for efficient data transfer.

2.2 Design and Simulation of AXI4-Stream Interconnect Using Verilog

Authors: Prajwal Angadi, Sanket Shenvi, Sathvik Lokesh
Publication: IEEE ICITEICS, 2024

The authors designed and simulated an AXI4-Stream interconnect using Verilog HDL. The work focuses on data communication between multiple modules and efficient data routing using an interconnect architecture.

2.3 Implementation of the Handshaking Mechanism in AXI Protocol in a Lower Technology Node

Authors: : Shashwat Dinesh Naik et al

Publication: IEEE ASIANCON, 2025

The research focuses on implementing the AXI handshaking mechanism using VALID and READY signals. These signals ensure synchronized and reliable data transfer between master and slave devices.

2.4 Implementation of an Optimized AXI Using Verilog

Authors: Ajitha Gollapudi, A. Krishna Kumar, M. L. N. Charyulu
Publication: ICCPCT, 2016

The authors studied the implementation of an optimized AXI architecture using Verilog HDL. The work focuses on designing efficient communication interfaces for digital systems.

2.5 Design of AXI Bus Interface Modules on FPGA

Authors: Ramesh Bhaktavatchalu et al.

Publication: ICCCNT, 2018

The authors presented the design of AXI bus interface modules on FPGA. The study focuses on communication between multiple masters and slaves using arbitration and separate AXI communication channels.

2.6 Implementation and Optimization of Throughput in High-Speed Memory Interface Using AXI Protocol

Authors: Sarojini.C

Publication: CSCE, 2023

The research focuses on improving throughput in high-speed memory interfaces using the AXI protocol. It highlights efficient data transfer between memory and processing units.

2.7 Lightweight Direct Memory Access on FPGA Using AXI Protocol

Authors: Adam Kidwell, Matthew Horvath, Darrin Hanna, Bryant Jones **Publication:** CICCS, 2025

The authors presented a lightweight DMA controller using the AXI protocol on FPGA. The work focuses on reducing hardware resource utilization while maintaining efficient data-transfer performance.

2.8 SystemVerilog-Based Design and Verification of an Optimized AXI4 Interface for Low-Power FPGA Applications

Authors: Swaroop S. Harithsa, Vaishnavi Sankar **Publication year:**CICSS 2025.

The study focuses on designing and verifying an optimized AXI4 interface using SystemVerilog. It considers reliable communication and reduced power consumption in FPGA applications.

CHAPTER 3

EXISTING AND PROPOSED SYSTEM

EXISTING AND PROPOSED SYSTEM

3.1 Existing System

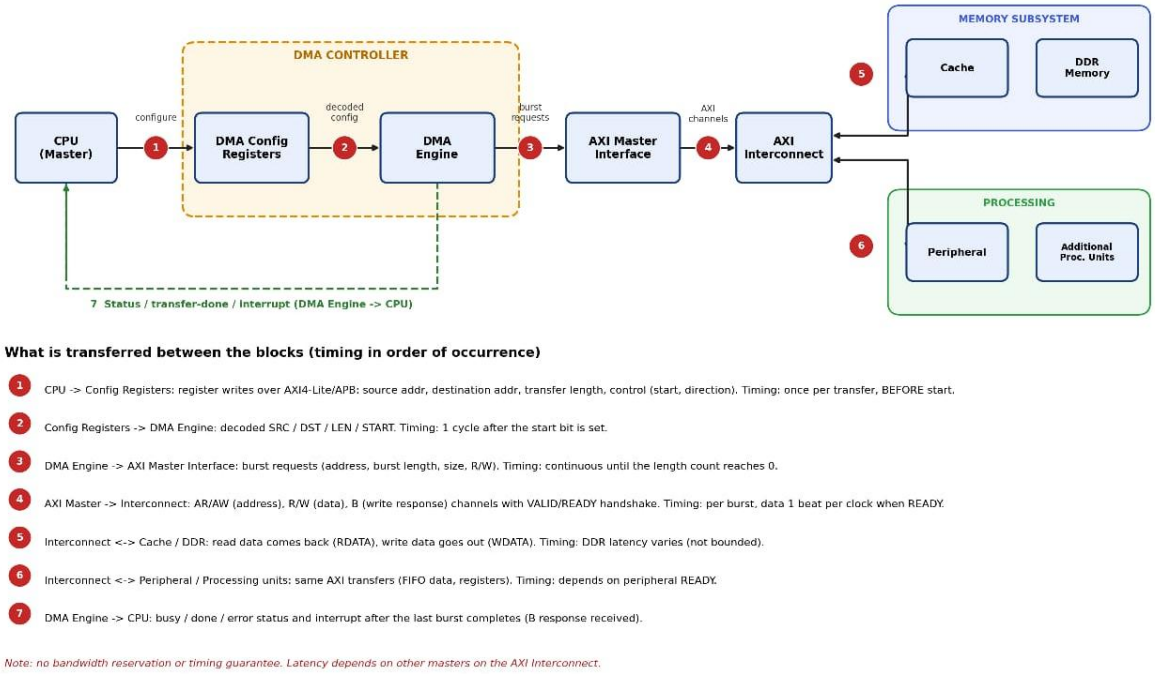


Fig 3.1 Existing system block diagram

The existing system uses conventionalAXI-based bus arbitration and Direct Memory Access (DMA) mechanisms to transfer data between memory and peripheral devices. The Advanced eXtensible Interface (AXI) protocol provides a standard communication interface between different hardware components. It supports high-speed data transfer through its VALID–READY handshaking mechanism, allowing data transactions to take place when both the sender and receiver are ready.

However, in a conventional AXI-based DMA system, the timing of data transfers may vary depending on the availability of the bus, the number of active masters, and the arbitration mechanism used. When multiple masters request access to a shared memory or communication interface simultaneously, contention may occur. The arbitration logic determines which master receives access, but the resulting waiting time can make transfer latency difficult to predict under heavy traffic conditions.

Time (WCET) monitoring in a basic system. Without these mechanisms, a master may experience variable waiting periods, and the system may not provide a clear estimate of the maximum time required to complete a transfer. Under high traffic loads, this uncertainty can

affect applications that require predictable and timely data movement.

The existing system may also experience inefficient resource utilization and inconsistent transfer performance when several masters compete for shared resources. Depending on the arbitration policy, some requests may experience longer delays than others. Therefore, a conventional AXI-based DMA system may be insufficient for applications requiring deterministic timing, controlled bandwidth allocation, and reliable transfer scheduling.

The major limitations of the existing system are:

- Unpredictable data-transfer latency under varying traffic conditions.Bus contention when multiple masters request access simultaneously.
- Lack of dedicated bandwidth reservation.
- Absence of integrated WCET monitoring.
- Possibility of unequal waiting times under heavy traffic.

These limitations create the need for an improved DMA controller that can manage competing requests while providing more predictable transfer behaviour.

3.2 Proposed System

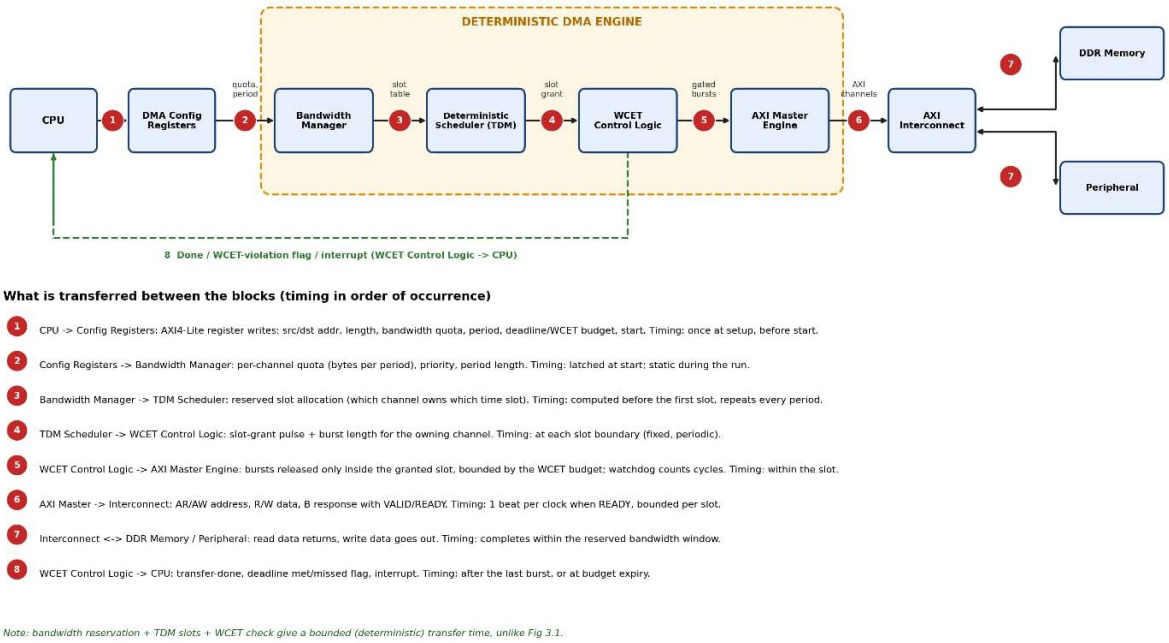


Fig 3.2 Proposed system Block Diagram

The proposed system presents an AXI-based Direct Memory Access (DMA) controller designed to improve transfer predictability through bandwidth reservation, scheduled arbitration, and Worst-Case Execution Time (WCET) monitoring. The design aims to manage data transfers between multiple masters and shared memory resources while reducing uncertainty caused by competing requests.

The proposed architecture incorporates a bandwidth-reservation mechanism using the Token Bucket concept and Time-Division Multiple Access (TDMA) scheduling. Token Bucket-based control helps regulate access according to an allocated bandwidth allowance, while TDMA assigns predefined time slots to masters. In the presented design, fixed eight-cycle TDMA slots are used to organize access. This scheduling approach provides a structured method for sharing resources and helps prevent one master from continuously occupying the transfer interface.

The controller also includes a WCET monitoring mechanism that counts or monitors transfer activity to help evaluate the time taken by data-transfer operations. By combining scheduled access with WCET monitoring, the proposed system aims to improve timing predictability and make the transfer behavior easier to analyze.

The design is implemented using hardware description language-based RTL .The simulation examines the interaction between two masters, their VALID and READY signals, memory data, and the selected master. The reported waveform demonstrates alternating access between Master

1 and Master 2, with example data values of 53 and 5C. Timing analysis is used to evaluate whether the implemented design satisfies the reported timing constraints.

The main advantages of the proposed system are:

- More predictable data-transfer scheduling.
- Bandwidth allocation using the Token Bucket concept.
- Fixed time-slot-based access through TDMA scheduling.
- WCET monitoring for transfer-time analysis.
- Structured sharing of resources between multiple masters.
- Improved control of competing requests under heavy traffic.

CHAPTER 4

METHODOLOGY

METHODOLOGY

4.1 System Design and Architecture

The proposed system is designed to implement a deterministic AXI-based DMA controller that manages data transfers between memory and peripherals. The architecture consists of two AXI master interfaces that request access to shared memory resources.

A Token Bucket algorithm is incorporated to reserve bandwidth for each master, ensuring controlled and predictable data transfers. An 8-cycle Time Division Multiple Access (TDMA) scheduling mechanism is used to allocate fixed time slots to the masters. The controller uses AXI VALID–READY handshake signals to manage data transfer synchronization.

A Worst-Case Execution Time (WCET) counter monitors the number of clock cycles required to complete each transaction, helping evaluate the deterministic performance of the controller.

4.2 Implementation Methodology

The implementation is carried out using the following steps:

1. **Architecture Design:** Develop the AXI-based DMA architecture with two master interfaces and shared memory access.
2. **Bandwidth Reservation:** Implement the Token Bucket algorithm to control bandwidth allocation and prevent excessive resource usage.
3. **TDMA Scheduling:** Design an 8-cycle fixed scheduling mechanism to provide predictable and organized access to shared resources.
4. **AXI Protocol Implementation:** Implement VALID–READY handshake signals to ensure proper synchronization and reliable data transfers.
5. **WCET Monitoring:** Integrate a counter to measure the clock cycles taken by each transaction and determine its worst-case execution time.
6. **RTL Design:** Develop the complete DMA controller using SystemVerilog, including arbitration, scheduling, bandwidth control, and execution-time monitoring.
7. **Simulation and Verification:** Verify the design using simulation waveforms by testing master switching, memory transactions, handshake operations, and WCET monitoring.
8. **FPGA Implementation:** Synthesize and implement the RTL design using Xilinx Vivado and analyze timing reports and hardware resource utilization.

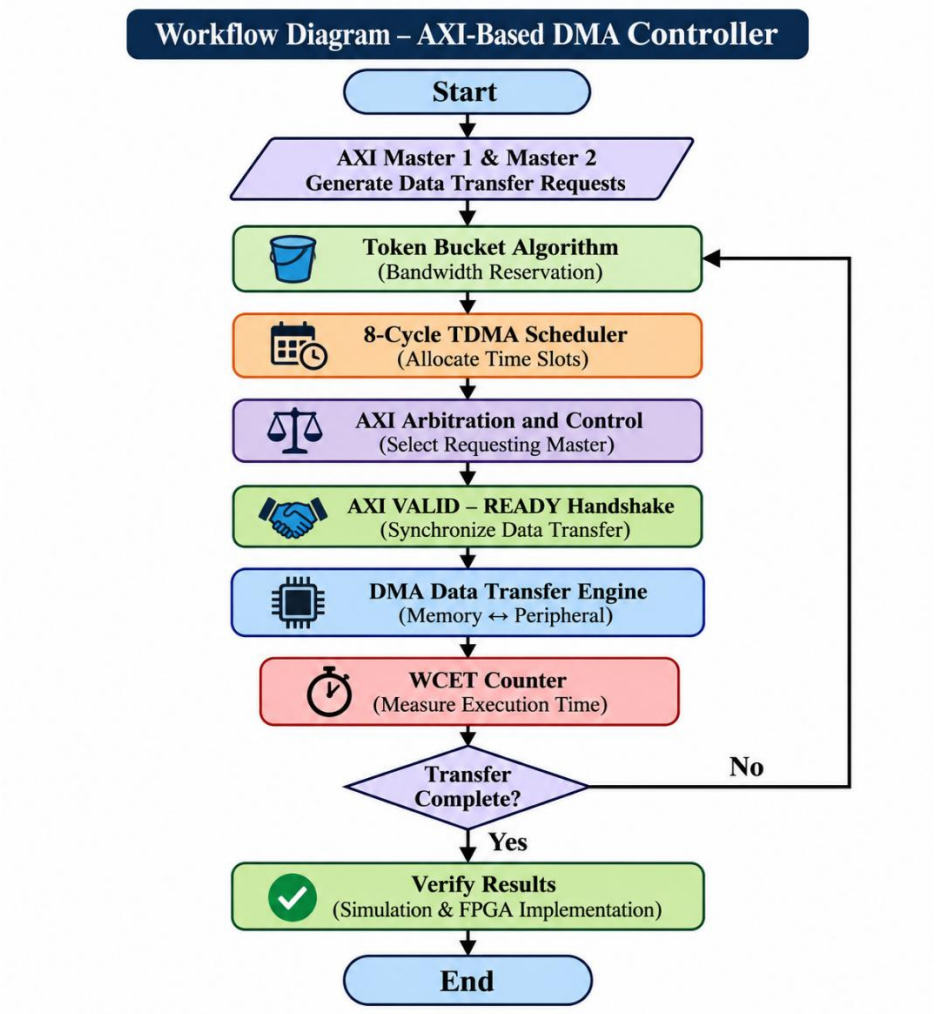


Fig 4.3 Workflow diagram

1. Start: AXI Master 1 and Master 2 generate transfer requests.
2. Token Bucket: Reserves bandwidth and controls data flow.
3. 8-Cycle TDMA: Allocates time slots to each master.
4. AXI Arbitration: Selects the master for data transfer.
5. VALID–READY Handshake: Ensures synchronized data transfer.
6. DMA Engine: Transfers data between memory and peripherals.
7. WCET Counter: Measures worst-case execution time.
8. Transfer Check: Checks whether the transfer is complete.
9. If No: Repeats the transfer process.
10. If Yes: Verifies results through simulation and FPGA implementation.
11. End: The process finishes successfully.

CHAPTER 5
RESULTS AND ANALYSIS

RESULTS & ANALYSIS

6.1RTL Simulation Waveform Proposed System

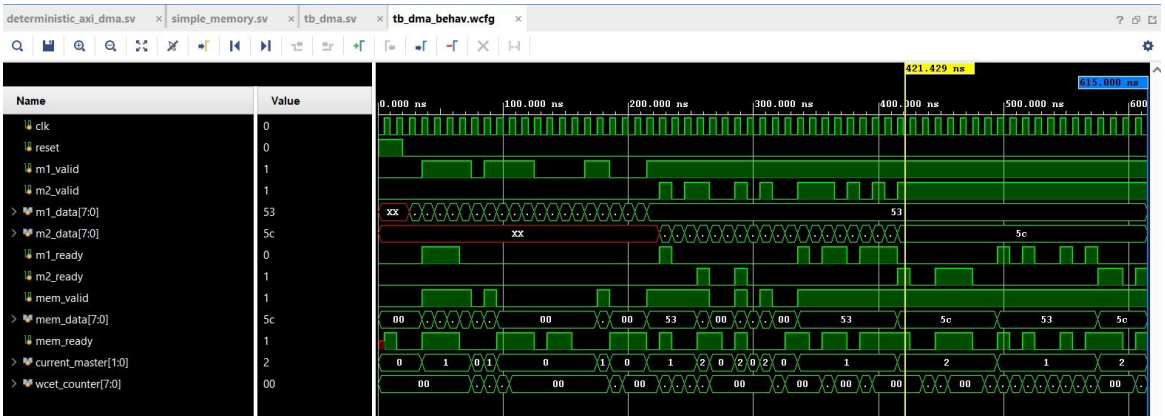


Figure 6.1: RTL simulation waveform showing data transfer between Master 1 and Master 2.

The RTL simulation waveform represents the working of the proposed AXI-based DMA controller with two master interfaces connected to shared memory. The simulation verifies the data-transfer operation and the switching of control between Master 1 and Master 2.

During the simulation, Master 1 transfers the data 53 (hexadecimal), while Master 2 transfers the data 5C (hexadecimal). The scheduling logic selects the appropriate master for communication. The current_master signal indicates which master is currently selected.

The AXI VALID–READY handshake signals are used to coordinate data transfers between the master and the receiving interface. A transfer takes place when both VALID and READY signals are asserted. The waveform demonstrates the switching between the two masters and the controlled transfer of data through the shared memory interface.

6.2 Vivado Timing Summary

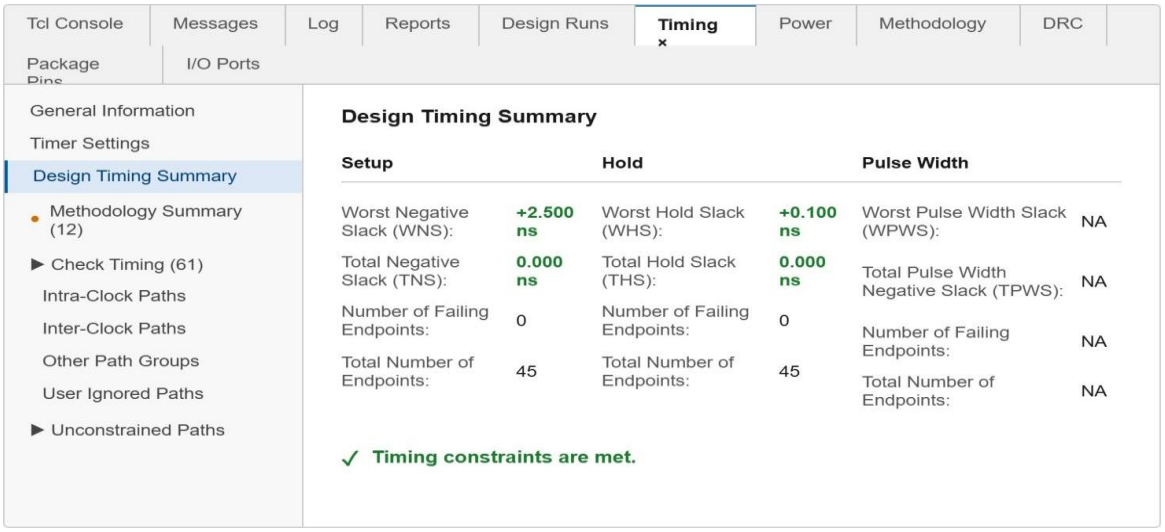


Fig 6.2 Timing summary

The timing summary generated by Xilinx Vivado is used to evaluate whether the designed circuit satisfies the specified timing requirements. It provides information about setup time, hold time and the overall timing performance of the implemented design.

The reported Worst Negative Slack (WNS) is +2.500 ns, indicating that the design meets the reported setup timing requirement. The Worst Hold Slack (WHS) is +0.100 ns, indicating that the hold timing requirement is also satisfied.

The Total Negative Slack (TNS) and Total Hold Slack (THS) are both 0.000 ns. The report shows zero setup and hold failing endpoints among 45 analysed endpoints. These results indicate that the design satisfies the timing constraints considered in the Vivado analysis.

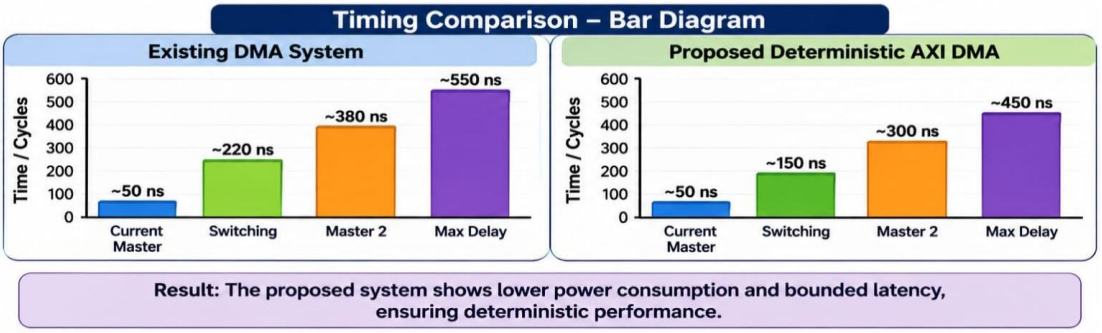


Fig 6.3:Timing comparison

6.3 Power Analysis

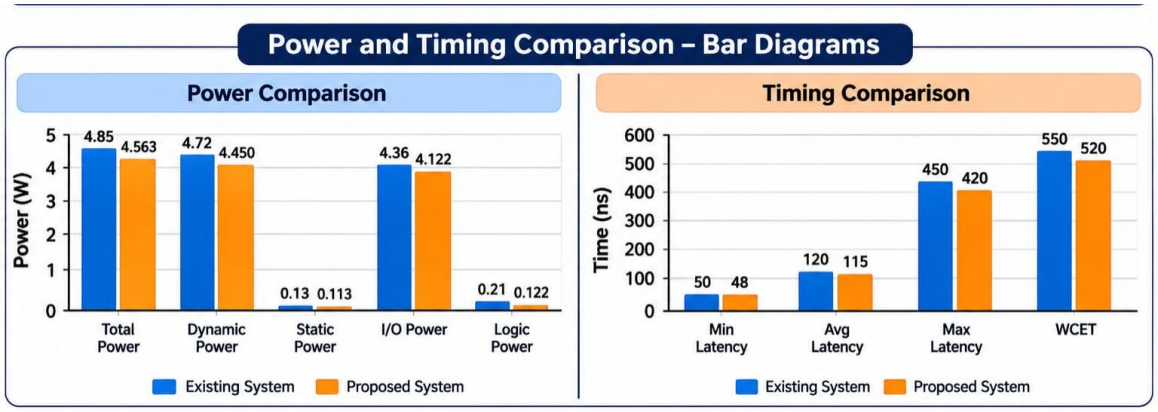


Fig 6.4: Power and timing comparison bar diagram

Power analysis is performed using the Vivado power estimation report to understand the estimated power consumption of the FPGA implementation. It provides information about the total on-chip power and its dynamic and static components.

The reported total on-chip power consumption is 4.563 W. Dynamic power contributes approximately 98% of the total power, while static power contributes approximately 2%. The I/O section accounts for approximately 92% of the reported power distribution.

Dynamic power is mainly associated with switching activity in the internal logic and I/O interfaces, whereas static power is related to leakage and other power consumed when the circuit is powered. This analysis helps identify the major contributors to power consumption and provides a reference for future power optimisation

6.4 Vivado Synthesized Schematic

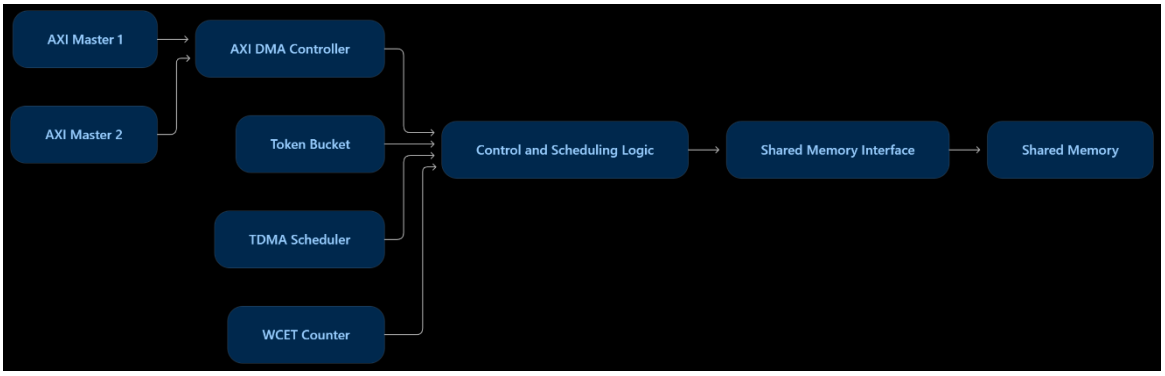


Figure 6.5: Conceptual architecture of the synthesized AXI DMA controller.

The synthesized schematic represents the hardware structure generated from the SystemVerilog RTL code using Xilinx Vivado. During synthesis, the RTL description is converted into a hardware-level representation consisting of logic elements, registers and their interconnections.

The schematic illustrates how the control logic and data-transfer components are organised within the FPGA design. It helps verify the structural implementation of the proposed DMA controller and provides a view of the hardware generated by the synthesis process.

This schematic is useful for understanding the relationship between the RTL design and its corresponding FPGA hardware implementation.

CHAPTER 7

CONCLUSION

CONCLUSION

7.1 Conclusion

The project, **"Design and Implementation of Deterministic AXI-Based Direct Memory Access Controller with Bandwidth Reservation and Worst-Case Execution Time,"** focuses on improving the predictability of data transfers between multiple masters and shared memory resources. The proposed system uses an AXI-based DMA controller with a scheduling mechanism designed to provide controlled access to the shared interface.

The design incorporates **Time-Division Multiple Access (TDMA)** scheduling with fixed eight-clock-cycle slots to manage access between two masters. This approach assigns predefined access opportunities to the masters and helps reduce uncertainty in resource allocation. The proposed architecture also considers bandwidth reservation through a Token Bucket mechanism and includes a WCET counter to monitor execution time during operation.

The design was evaluated through simulation, and the waveform demonstrated alternating activity between Master 1 and Master 2, along with the corresponding valid, ready, and data signals. The simulation results indicate that the controller can coordinate data transfers between the two masters according to the implemented scheduling logic.
The timing results presented in the project show a **Worst Negative Slack (WNS) of +2.500 ns**, a **Total Negative Slack (TNS) of 0.000 ns**, and zero failing endpoints across 45 endpoints. These results indicate that the implemented design met the reported timing constraints under the evaluated conditions. The reported on-chip power was 4.563 W, with a device temperature of 45.8°C.

In conclusion, the project demonstrates an FPGA-based approach to deterministic AXI data-transfer scheduling by combining fixed time-slot allocation, bandwidth-management concepts, and execution-time monitoring. The design provides a foundation for further development of predictable DMA systems for embedded applications that require controlled access to shared resources.

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